

Matched stretch: what the runs show

The simulations show both decay and sustained spin amplification from smooth initial data, but resolution dependence prevents a conclusion about singularity or global regularity.

Definitions

Speed means the largest velocity magnitude, $\max |u|$. Biggest spin, written $W_{\max}(t)$, means the largest vorticity magnitude, $\max |\text{curl } u|$. The spin ratio is $W_{\max}$ divided by the root-mean-square vorticity. The half-peak region is the set of cells where $|\text{curl } u|$ is at least half of $W_{\max}$. A fixed cutoff is a set level, such as 50 or 100, that does not move when the peak moves. The integral $I(t)$ is the time integral of $W_{\max}$ from 0 to t . The cube has side 6. Viscosity ν is 0.01 on the fading ring runs and 0.001 on the matched stretch.

Results

Fading starts. A ring at ν 0.01 has speed 0.06 at time 40. A tighter ring on 64 cubed falls from peak speed 5.47 to 0.86 by time 2. A pulled tube at ν 0.002 rises from 3.74 to 5.22, then falls to 1.46 by time 4. The same tube on 64 cubed rises from 3.60 to 6.16, then falls below the start. Two opposite tubes fall from 2.98 to 2.19.

Climbing start. A tight tube in a stretch, ν 0.001, periodic cube, no added force. On 48 cubed, $W_{\max}$ goes from 60 to 611 and is 514 at time 0.35. $I(0.36)$ is 119.9 and was still rising.

Same start, finer grids, at time 0.12. $W_{\max}$ is 194 on 48 cubed, 200 on 64 cubed, and 234 on 80 cubed. On 48 cubed the half-peak width falls from 30 cells to 6.6 cells, then rises to 11.6 cells by time 0.36.

Saved 48-cubed fields. Above a fixed spin of 50, the cell count is 930, then 15,815, then 40,524 at times 0, 0.12, and 0.24. Above 100 it is 0, then 1,210, then 11,013. Above 200 it is 0, then 0, then 1,342. Above 400 it is 11 cells at time 0.24. The half-peak cell count goes 27,191, then 2,146, then 485.

Pressure at the fastest point on the pulled tube switches sign. Smoothing there is smaller than the steepening at the samples taken. On the pulled tube the spin ratio falls. On one other start it rises from 30.6 to 32.2, then falls.

Interpretation

Decay and amplification both occur in these runs. That is an observation about the starts that were computed, not a statement about every smooth start.

The peak at time 0.12 is larger on the finer grid. The observation is the three numbers. The interpretation is that this transient has not converged. The half-peak region is still several cells wide, so the run has not collapsed to one cell. A moving cutoff and a fixed cutoff describe different regions: the core can tighten while more of the cube exceeds a fixed spin.

$I(0.36) = 119.9$ is finite, so the conditional smoothness criterion holds on this interval for this run. It does not say $I(t)$ stays finite later, and it does not say that for every start.

Passing time 0.0166 while remaining finite at time 0.35 rules out that particular crude estimate for this run. It does not constrain a later blowup, and it does not establish regularity.

Numerical controls

What is the same across the matched-stretch grids: the initial formula, $\nu = 0.001$, the periodic cube of side 6, a Fourier step with $2/3$ dealiasing, a pressure projection, and a time step tied to the cell width and the starting speed. Incompressibility is enforced by the projection. An energy decline was recorded on the pulled-tube runs, not on the matched stretch.

What was not done. No timestep was repeated at half size on the matched stretch. No second discretization was run. Grids of 112 cubed and 160 cubed were not run. The 64-cubed continuation past time 0.16 is unfinished. Those checks are required before the rising peak can be treated as a physical feature rather than a numerical one.